

# Oracle Database Administration

User Management, Schema Creation, Roles, Synonyms & Privilege Delegation Lab Report

## 1. Schema Initialization & User Creation ("SHAADMIN")

At first, create user **SHAADMIN** with specific tablespace quota. Under **SHAADMIN**, create three core relational tables: **EMP**, **DEPT**, and **LOC**.

### Create User SQL Query:

```
CREATE USER SHAADMIN IDENTIFIED BY ORACLE DEFAULT TABLESPACE USERS QUOTA 2G ON USERS;
```

```
SQL> CREATE USER SHAADMIN IDENTIFIED BY ORACLE DEFAULT TABLESPACE USERS QUOTA 2G ON USERS;
User created.
```

### Table Definitions under SHAADMIN Schema:

#### 1. EMP Table Creation:

```
CREATE TABLE EMP (EMP_ID NUMBER, NAME VARCHAR2(20), AGE NUMBER);
```

```
SQL> CREATE TABLE EMP (EMP_ID NUMBER, NAME VARCHAR2(20), AGE NUMBER);
Table created.
```

```
SQL> DESC EMP
Name                               Null?    Type
-----
EMP_ID                             NUMBER
NAME                               VARCHAR2(20)
AGE                                NUMBER
```

#### 2. DEPT Table Creation:

```
CREATE TABLE DEPT (DEPT_ID VARCHAR2(12), DEPT_NAME VARCHAR2(15), DEPT_NUMBER NUMBER);
```

```
SQL> CREATE TABLE DEPT (DEPT_ID varchar2(12), DEPT_NAME VARCHAR2(15), DEPT_NUMBER NUMBER);
Table created.
```

```
SQL> DESC DEPT;
Name                               Null?    Type
-----
DEPT_ID                             VARCHAR2(12)
DEPT_NAME                           VARCHAR2(15)
DEPT_NUMBER                          NUMBER
```

#### 3. LOC Table Creation:

```
CREATE TABLE LOC (LOC_ID VARCHAR2(12), LOC_NAME VARCHAR2(15), STATE VARCHAR2(15));
```

```
SQL> CREATE TABLE LOC (LOC_ID VARCHAR2(12), LOC_NAME VARCHAR2(15), STATE VARCHAR2(15));
Table created.

SQL> DESC LOC;
Name                               Null?    Type
-----
LOC_ID                             VARCHAR2(12)
LOC_NAME                           VARCHAR2(15)
STATE                              VARCHAR2(15)
```

## 2. DML Permissions & Data Population

Granting explicit DML permissions (SELECT, INSERT, UPDATE, DELETE) on existing schema tables before populating them with record data.

### Granting DML Permissions:

```
GRANT SELECT, INSERT, UPDATE, DELETE ON SHAADMIN.EMP TO SHAADMIN;
GRANT SELECT, INSERT, UPDATE, DELETE ON SHAADMIN.DEPT TO SHAADMIN;
GRANT SELECT, INSERT, UPDATE, DELETE ON SHAADMIN.LOC TO SHAADMIN;
```

```
SQL> GRANT SELECT,INSERT,UPDATE,DELETE ON SHAADMIN.EMP TO SHAADMIN;
Grant succeeded.

SQL> GRANT SELECT,INSERT,UPDATE,DELETE ON SHAADMIN.DEPT TO SHAADMIN;
Grant succeeded.

SQL> GRANT SELECT,INSERT,UPDATE,DELETE ON SHAADMIN.LOC TO SHAADMIN;
Grant succeeded.
```

### Populating and Verifying Table Records:

Populating table records and validating row insertions across tables.

```
SQL> SELECT * FROM EMP;

EMP_ID NAME                AGE
-----
1 ANIRBAN                  28
2 BILTU                    24
3 BRISHALI                 33
4 SUKRITI                  48
5 GITA                     36
6 MADHU                    22
7 JOLLY                    32
8 UMAN                     25

8 rows selected.
```

```
SQL>
SQL> SELECT * FROM DEPT;

DEPT_ID DEPT_NAME    DEPT_NUMBER
-----
D01     IT             101
D02     ANALYST        102
D03     FITTER          103
D04     CLERK           104
D05     DRIVER          105
D06     SALES           106
D07     STENO           107
D08     CIVIL           108

8 rows selected.
```

```
SQL> SELECT * FROM LOC;
```

| LOC_ID | LOC_NAME  | STATE          |
|--------|-----------|----------------|
| L01    | KOLKATA   | WEST BENGAL    |
| L02    | KOLKATA   | WEST BENGAL    |
| L03    | Mumbai    | Maharashtra    |
| L04    | Patna     | Bihar          |
| L05    | Lucknow   | Uttar Pradesh  |
| L06    | Bengaluru | Karnataka      |
| L07    | Jaipur    | Rajasthan      |
| L08    | Bhopal    | Madhya Pradesh |

```
8 rows selected.
```

### 3. Second User ("SHAUSER") & Public Synonyms

Creating user **SHAUSER** and establishing public synonyms for seamless database object access across different schema contexts.

```
CREATE USER SHAUSER IDENTIFIED BY ORACLE DEFAULT TABLESPACE USERS QUOTA 2G ON USERS;
```

```
SQL> CREATE USER SHAUSER IDENTIFIED BY ORACLE DEFAULT TABLESPACE USERS QUOTA 2G ON USERS;
```

```
User created.
```

```
SQL> COL USERNAME FOR A20;
```

```
SQL>
```

```
SQL> SELECT USERNAME FROM DBA_USERS WHERE USERNAME='SHAUSER';
```

```
USERNAME
```

```
-----  
SHAUSER
```

#### Public Synonym Creation:

```
CREATE PUBLIC SYNONYM EMPLOYEES FOR SHAADMIN.EMP;  
CREATE PUBLIC SYNONYM DEPARTMENTS FOR SHAADMIN.DEPT;  
CREATE PUBLIC SYNONYM LOCATIONS FOR SHAADMIN.LOC;
```

```
SQL> CREATE PUBLIC SYNONYM EMPLOYEES FOR SHAADMIN.EMP;
```

```
Synonym created.
```

```
SQL> CREATE PUBLIC SYNONYM DEPARTMENTS FOR SHAADMIN.DEPT;
```

```
Synonym created.
```

```
SQL> CREATE PUBLIC SYNONYM LOCATIONS FOR SHAADMIN.LOC;
```

```
Synonym created.
```

### 4. Role-Based Access Control (RBAC) & Privilege Assignment

Configuring a unified database role **SHA\_ROLE** to manage table permissions efficiently and granting it to **SHAUSER**.

## Create Role & Assign Table Permissions:

```
CREATE ROLE SHA_ROLE;
```

```
SQL> CREATE ROLE SHA_ROLE;  
Role created.
```

```
GRANT SELECT, INSERT, UPDATE ON SHAADMIN.DEPT TO SHA_ROLE;  
GRANT SELECT, INSERT, UPDATE ON SHAADMIN.LOC TO SHA_ROLE;
```

```
SQL> GRANT SELECT,INSERT,UPDATE ON SHAADMIN.DEPT TO SHA_ROLE;  
Grant succeeded.  
SQL> GRANT SELECT,INSERT,UPDATE ON SHAADMIN.LOC TO SHA_ROLE;  
Grant succeeded.
```

## Assign Role to SHAUSER:

```
GRANT SHA_ROLE TO SHAUSER;
```

```
SQL> GRANT SHA_ROLE TO SHAUSER;  
Grant succeeded.
```

## 5. Verification & Testing User Access

Logging in as **SHAUSER** to verify query execution and access rights via public synonyms.

```
SQL> CONN  
Enter user-name: SHAUSER  
Enter password:  
Connected.  
SQL>  
SQL>  
SQL> SELECT * FROM EMPLOYEES;  
  
  EMP_ID NAME                AGE  
-----  
    1 ANIRBAN                 28  
    2 BILTU                  24  
    3 BRISHALI                33  
    4 SUKRITI                 48  
    5 GITA                    36  
    6 MADHU                   22  
    7 JOLLY                   32  
    8 UMAN                    25  
  
8 rows selected.
```

```
SQL> SELECT * FROM DEPARTMENTS;
```

| DEPT_ID | DEPT_NAME | DEPT_NUMBER |
|---------|-----------|-------------|
| D01     | IT        | 101         |
| D02     | ANALYST   | 102         |
| D03     | FITTER    | 103         |
| D04     | CLERK     | 104         |
| D05     | DRIVER    | 105         |
| D06     | SALES     | 106         |
| D07     | STENO     | 107         |
| D08     | CIVIL     | 108         |

8 rows selected.

```
SQL> SELECT * FROM LOCATIONS;
```

| LOC_ID | LOC_NAME  | STATE          |
|--------|-----------|----------------|
| L01    | KOLKATA   | WEST BENGAL    |
| L02    | KOLKATA   | WEST BENGAL    |
| L03    | Mumbai    | Maharashtra    |
| L04    | Patna     | Bihar          |
| L05    | Lucknow   | Uttar Pradesh  |
| L06    | Bengaluru | Karnataka      |
| L07    | Jaipur    | Rajasthan      |
| L08    | Bhopal    | Madhya Pradesh |

8 rows selected.

## 6. User Creation ("PWC") & Direct SELECT Privilege Grant

Creating an additional user **PWC** and granting direct SELECT permissions on specific schema objects.

```
CREATE USER PWC IDENTIFIED BY PWC DEFAULT TABLESPACE USERS QUOTA 1G ON USERS;  
GRANT SELECT ON SHAADMIN.EMP TO PWC;
```

```
SQL> conn
```

```
Enter user-name: pwc/pwc
```

```
Connected.
```

```
SQL>
```

```
SQL>
```

```
SQL> select * from employees;
```

| EMP_ID | NAME     | AGE |
|--------|----------|-----|
| 1      | ANIRBAN  | 28  |
| 2      | BILTU    | 24  |
| 3      | BRISHALI | 33  |
| 4      | SUKRITI  | 48  |
| 5      | GITA     | 36  |
| 6      | MADHU    | 22  |
| 7      | JOLLY    | 32  |
| 8      | UMAN     | 25  |

8 rows selected.